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Subject Name :Data Structures

Subject code:0316

ASSESSMENT TOLL-C02(AT-2)

1. Find the bugs, include the missing lines in the following snippet.

int linearSearch(int arr[], int n, int key) {

for(int i = 0; i <= n; i++) {

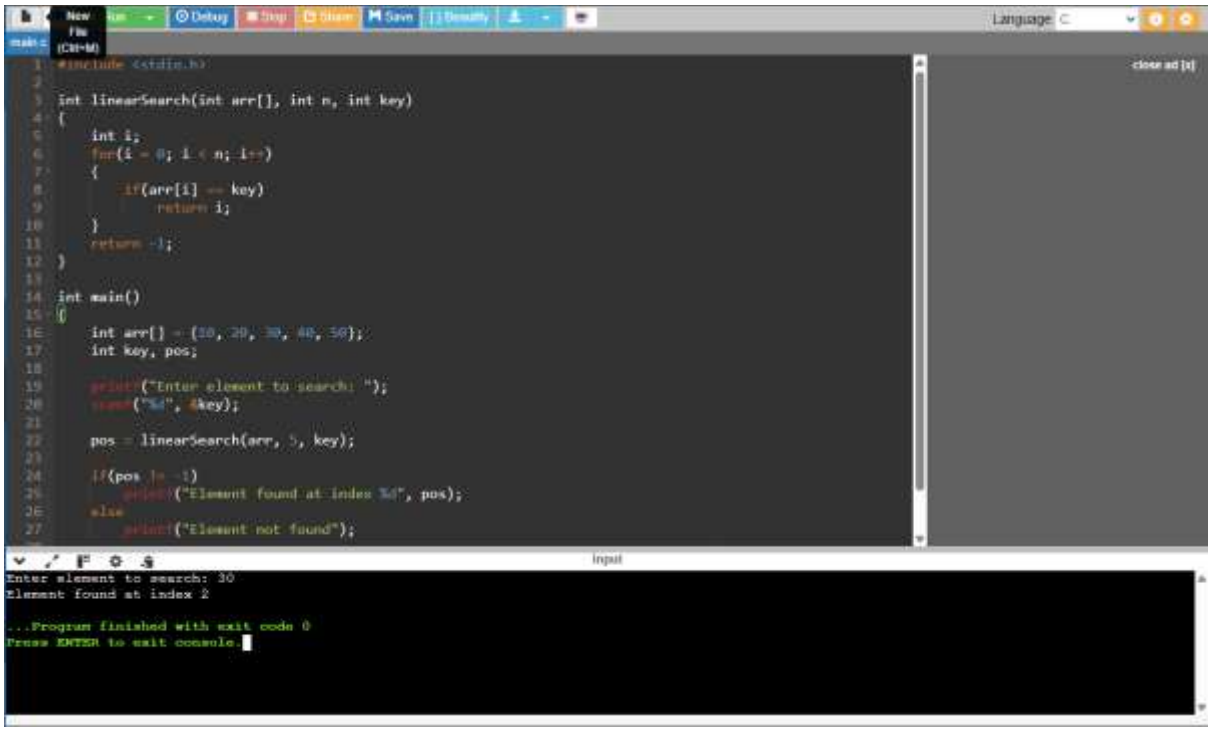
if(arr[i] == key)

return i;

}

return -1;

}



2. Buggy Code in an array

#include <stdio.h>

int main() {

int arr[5] = {10,20,30,40,50};

for(int i=0;i<=5;i++)

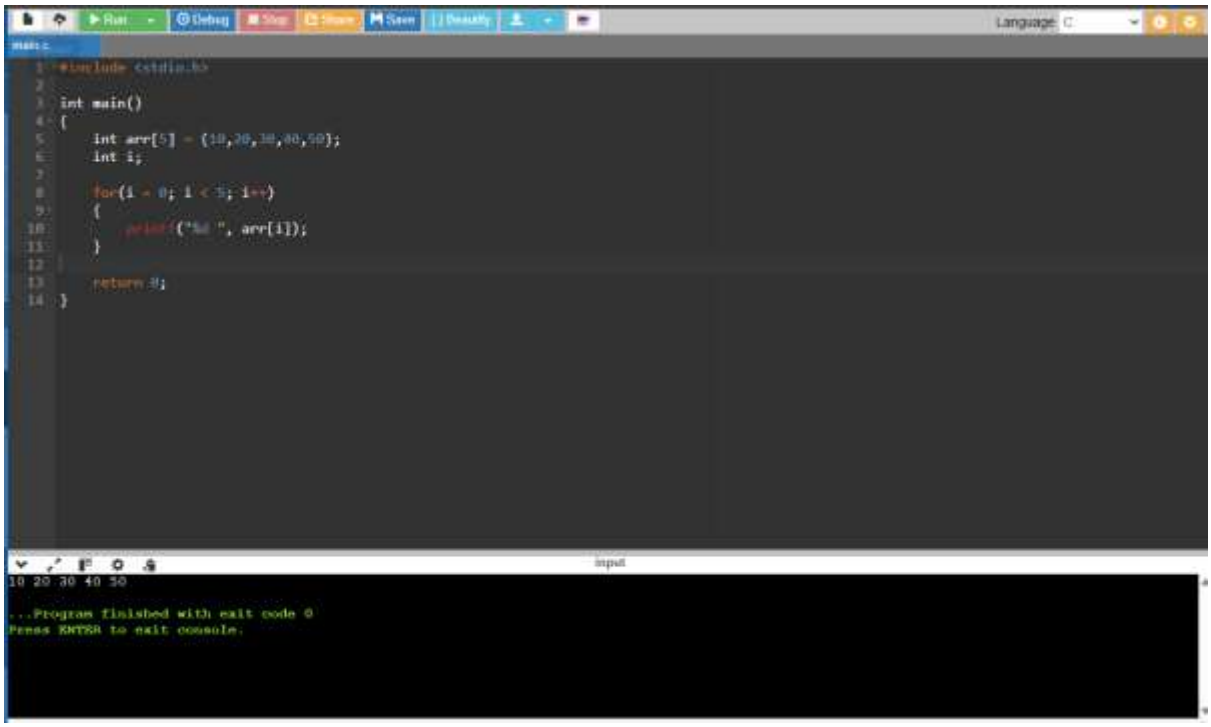
printf("%d ",arr[i]);

return 0;

}

Expected output:

10 0 30 40 50



3. Identify an errors in the following snippet

int binarySearch(int arr[], int n, int key)

{

int low = 0, high = n - 1;

while(low < high)

{ int mid = (low + high) / 2;

if(arr[mid] == key) return mid;

else if(arr[mid] < key)

low = mid;

else high = mid;

}

return -1;

}

4. Buggy Code in Stack Push Operation

#include <stdio.h>

#define MAX 5

int stack[MAX];

int top = -1;

void push(int data) {

if (top == MAX - 1)

printf("Stack Overflow");

else

stack[top++] = data;

}

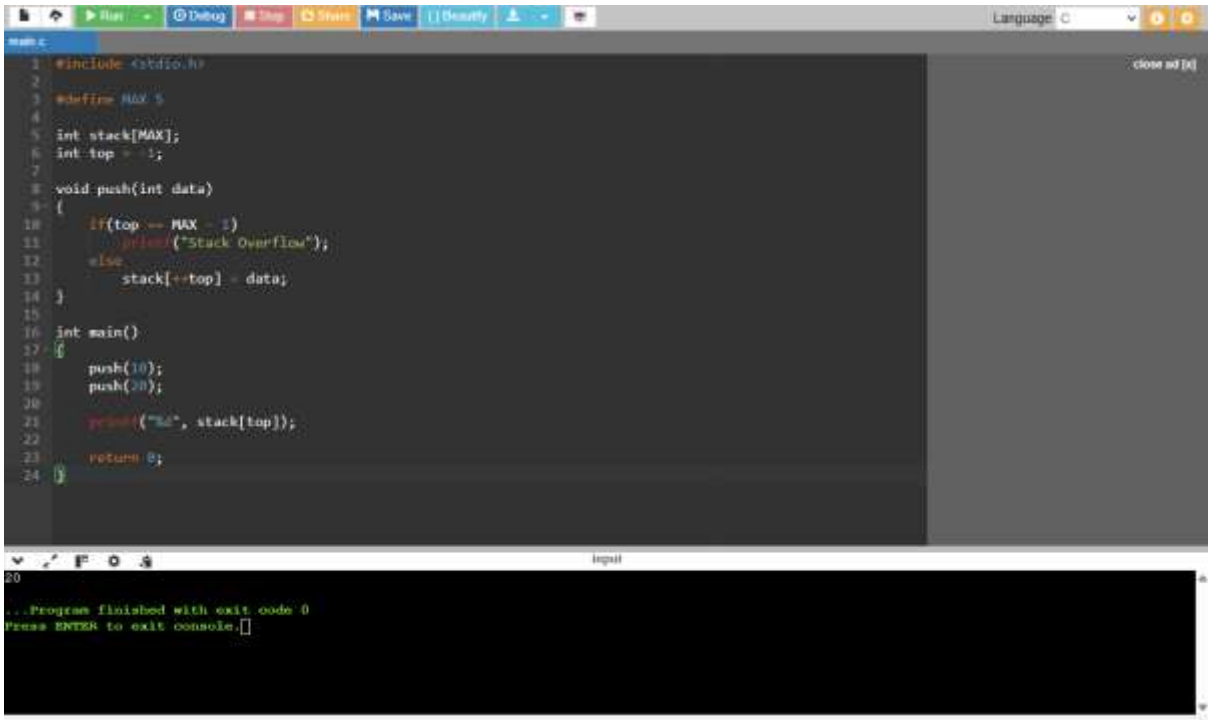
int main() {

push(10);

push(20);

printf("%d", stack[top]);

}



5. Buggy Code in Stack POP Operation

#include <stdio.h>

#define MAX 5

int stack[MAX];

int top = 2;

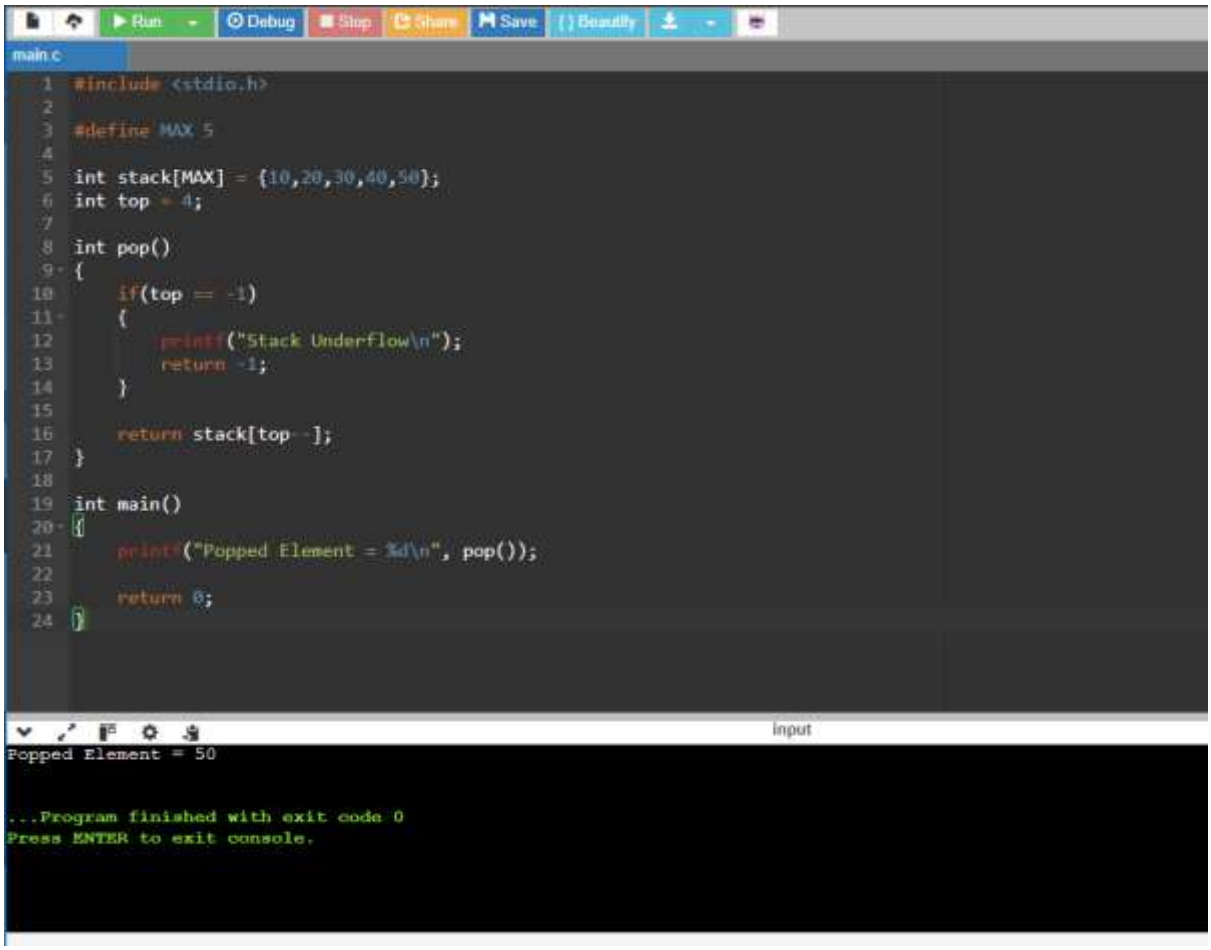
int pop() {

if (top == -1)

return -1;

return stack[++top];

}



6. Buggy Code in underflow operation

int pop() {

if(top == 0) {

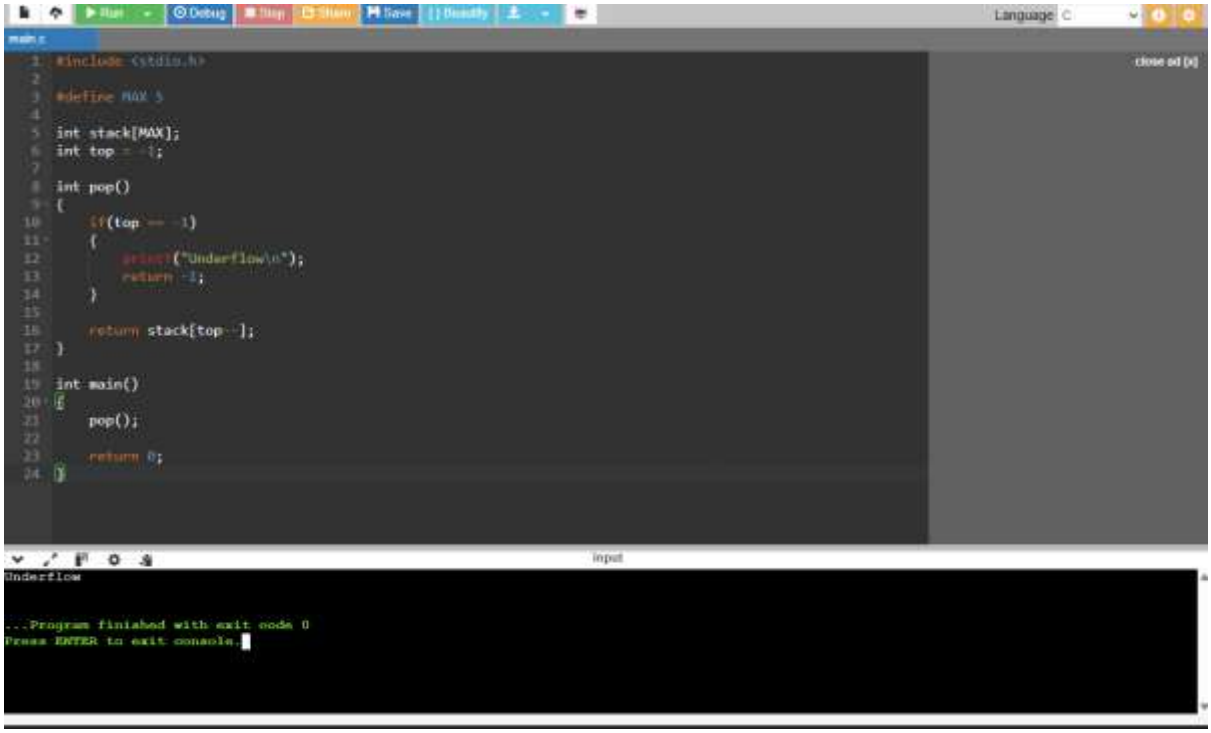
printf("Underflow");

return -1;

}

return stack[top--];

}



7. Find the issue in the following code snippet.

#include<stdio.h>

#define MAX 5

void main()

{

int queue[MAX],x;

int front = 0, rear = 0;

printf("Enter the value");

scanf("%d",&x);

void enqueue(int x) {

if(rear == MAX) {

printf("Overflow");

return;

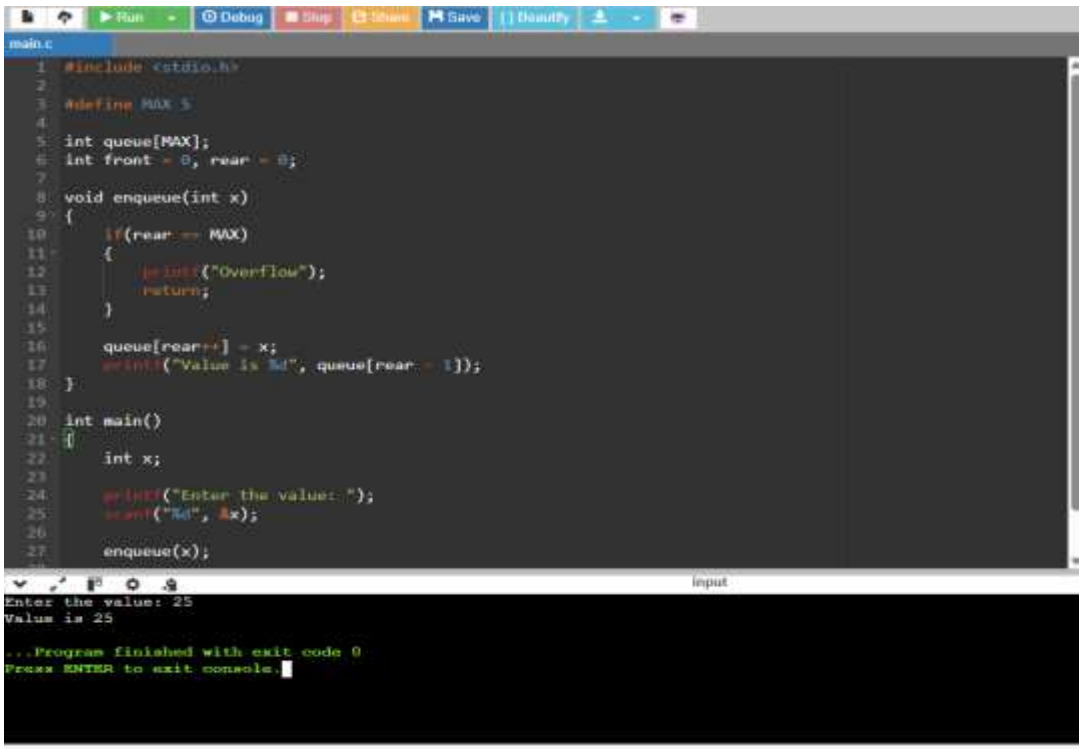
}

else{

queue[rear++] = x;

printf("Value is %d", queue[rear]);

}}}



8. Buggy code in node creation

#include <stdio.h>

#include <stdlib.h>

struct Node {

int data;

struct Node *next;

};

int main() {

struct Node *newNode;

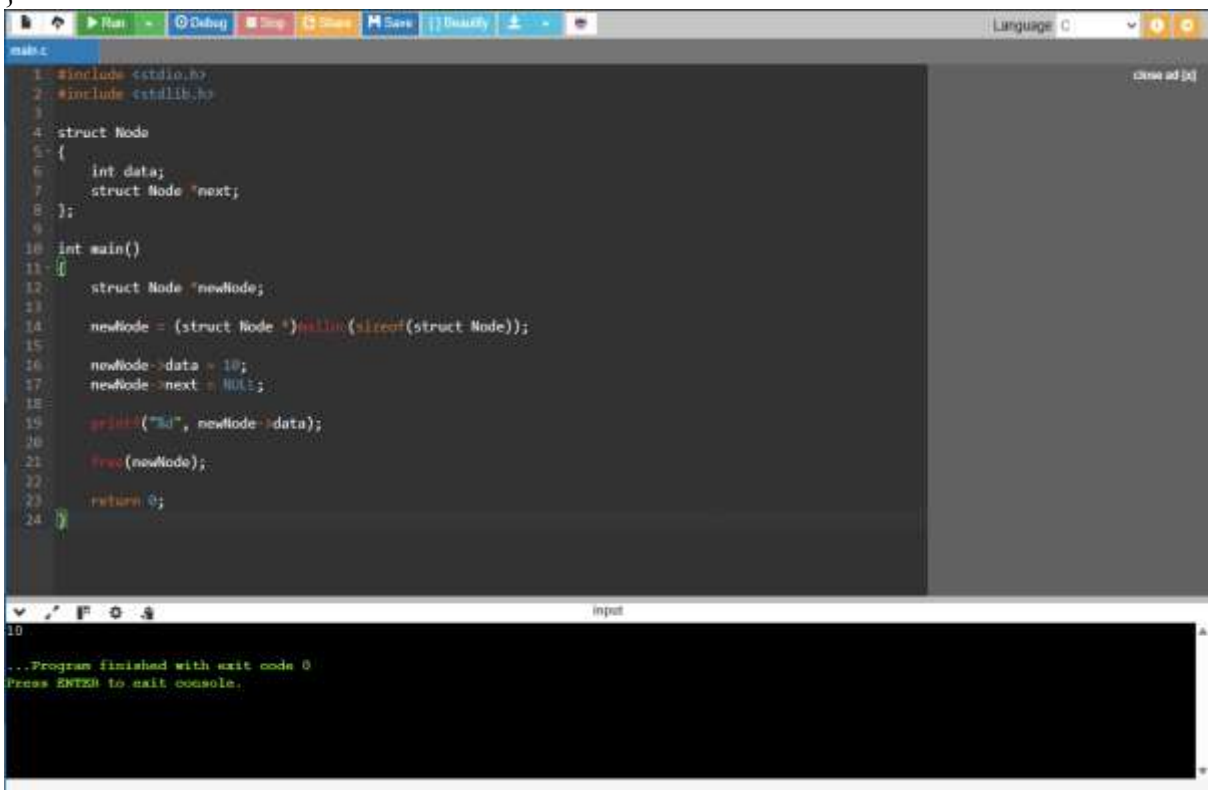
newNode->data = 10;

newNode->next = NULL;

printf("%d", newNode->data);

return 0;

}



9. Include missing statements and display the list of elements in linkedlist

void display(struct Node *head)

{

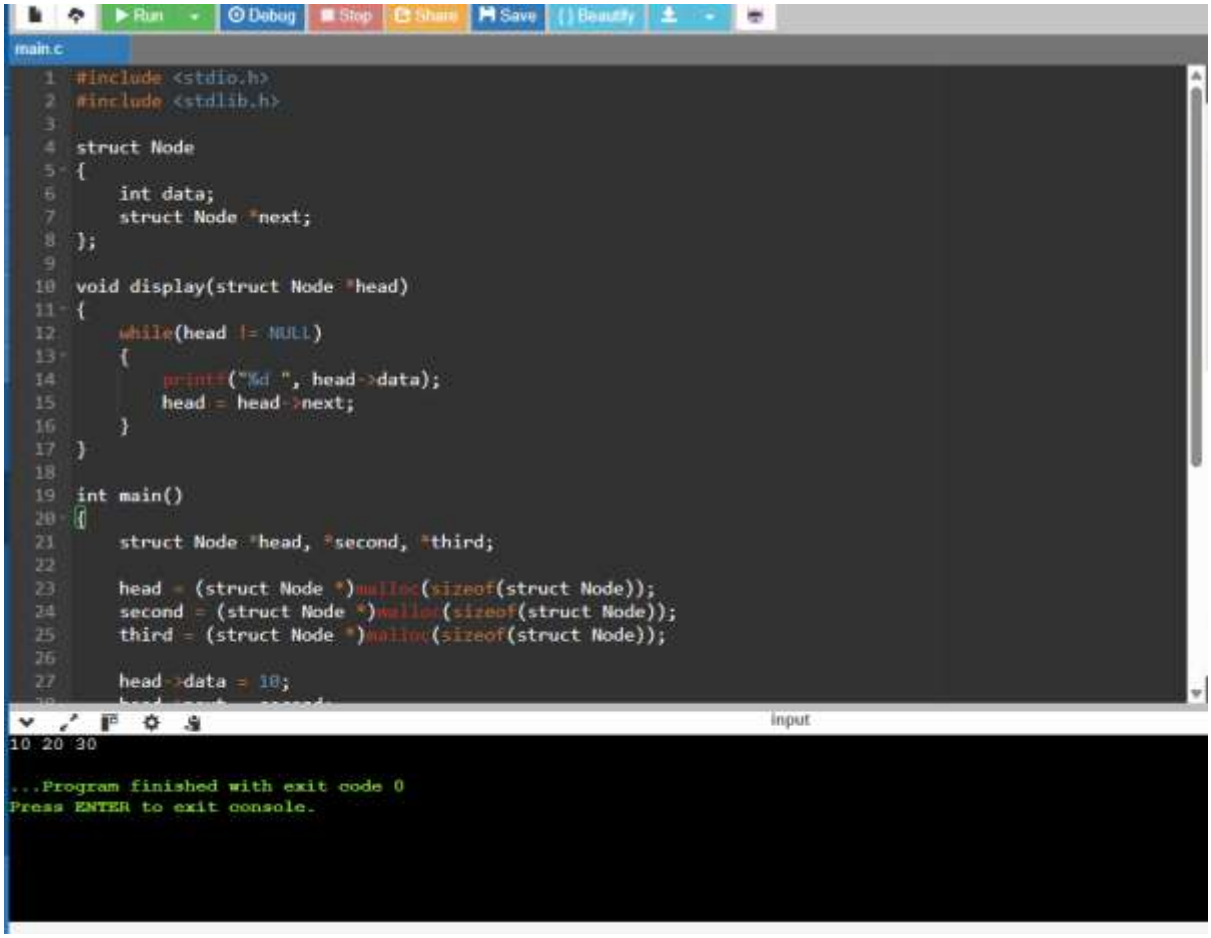
while(head != NULL);

{

printf("%d ", head->data);

head = head->next;

} }



10. Bug in Queue Full Condition-Identify and fix the errors.

#define SIZE 5

int queue[SIZE];

int front = -1, rear = -1;

void enqueue(int item)

{

if(rear == SIZE-1)

{

printf("Queue Full");

return;

}

if(front == -1)

front = 0;

rear = (rear + 1) % SIZE;